

Effects of urbanization on freshwater fish species community structure in the Etowah River watershed

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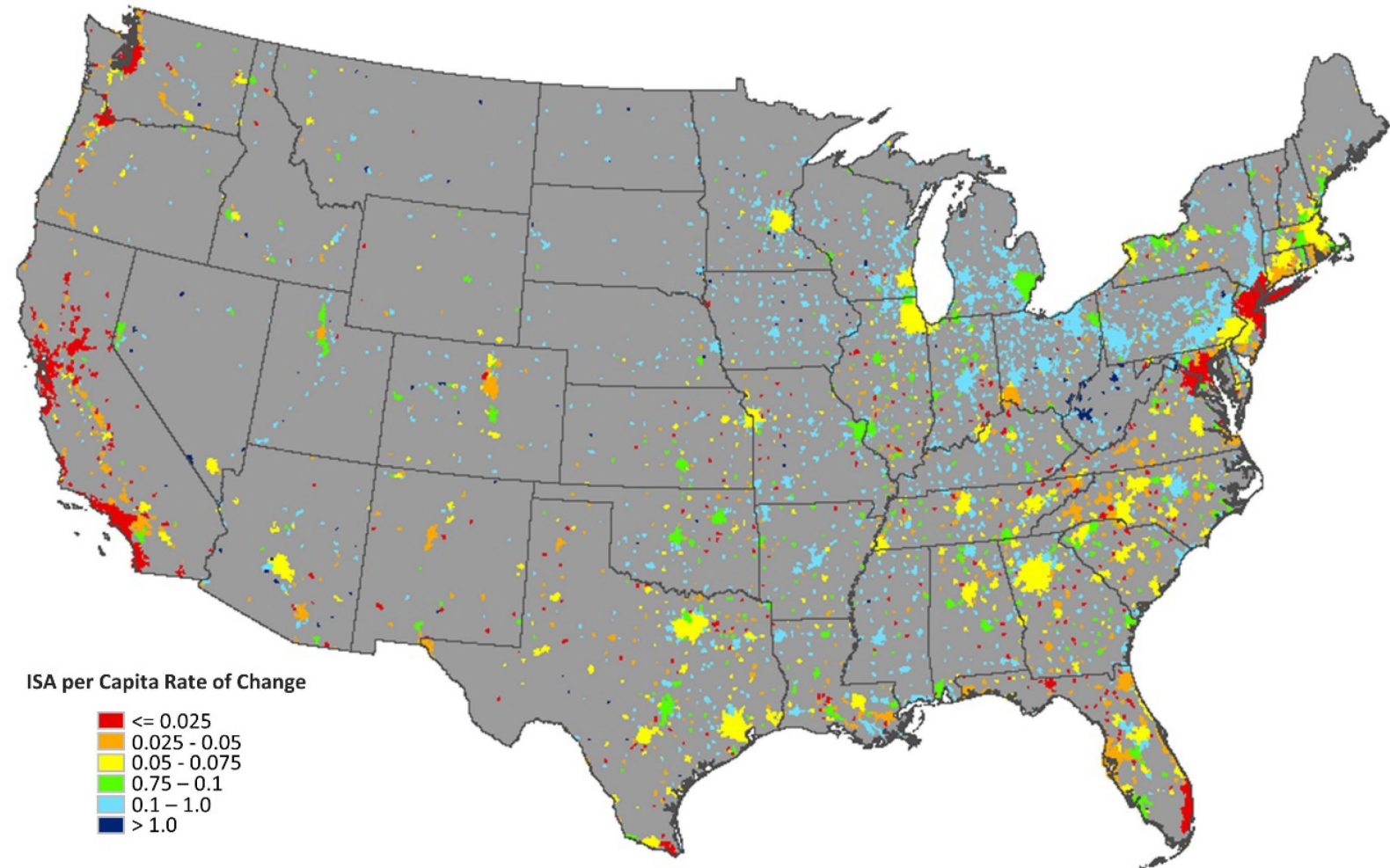


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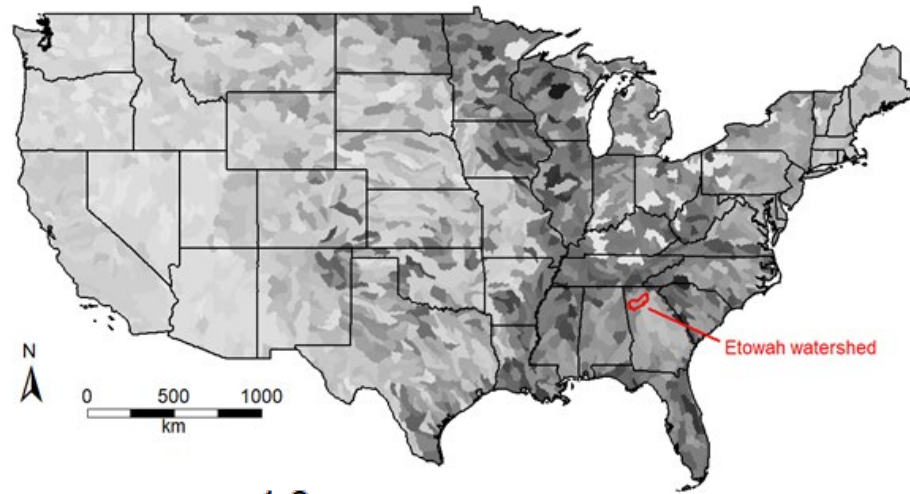


Urbanization

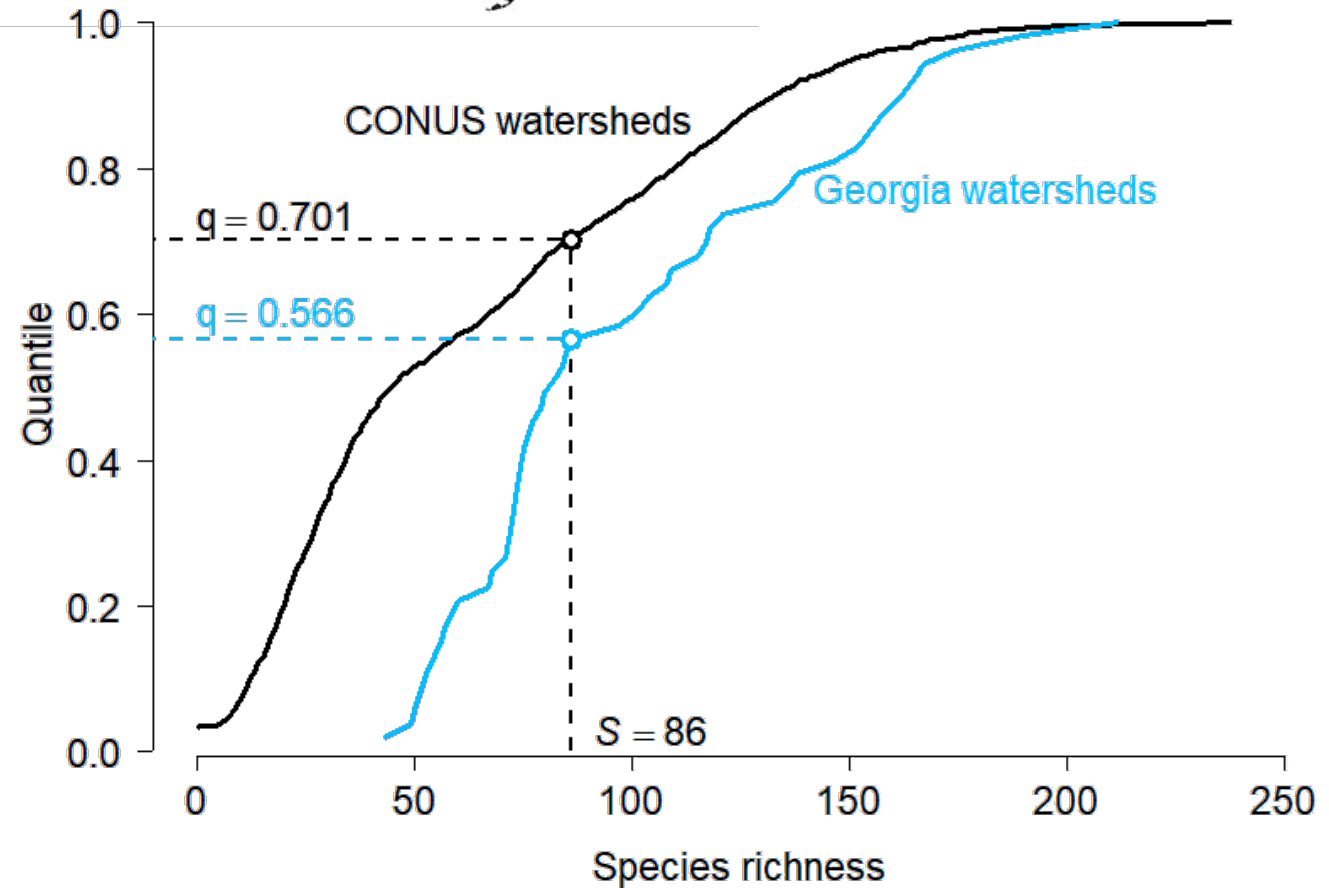
- Increase in impervious surface
 - Reduction in water quality
 - Increased surface runoff
- Decrease in forest cover
 - Increase in water temperature
 - Impacts food webs
- Reduce channel connectivity
- Stream fish communities reflect environmental conditions in the streams and in their watersheds



The Etowah Watershed



- 4791 km² in northern Georgia, with Lake Allatoona as a drinking reservoir
- Over 70 endemic fish species, at least 4 T&E species
- More fish diversity than 70% of continental US watersheds, 55% of Georgia watersheds
- Rapidly urbanizing due to Atlanta urban sprawl



Fish traits and life history strategies

- Species with similar life history traits may respond to environmental conditions in similar ways
- Possible use of traits as framework for predicting effects of climate and land cover change on fish communities and river ecosystems



Largescale stoneroller
Campostoma oligolepsis

benthopelagic
rocky substrates
nonguarder



Redbreast sunfish
Lepomis auritus

demersal
rocky and sandy substrates
guarder and nester



Redeye bass
Micropterus coosae

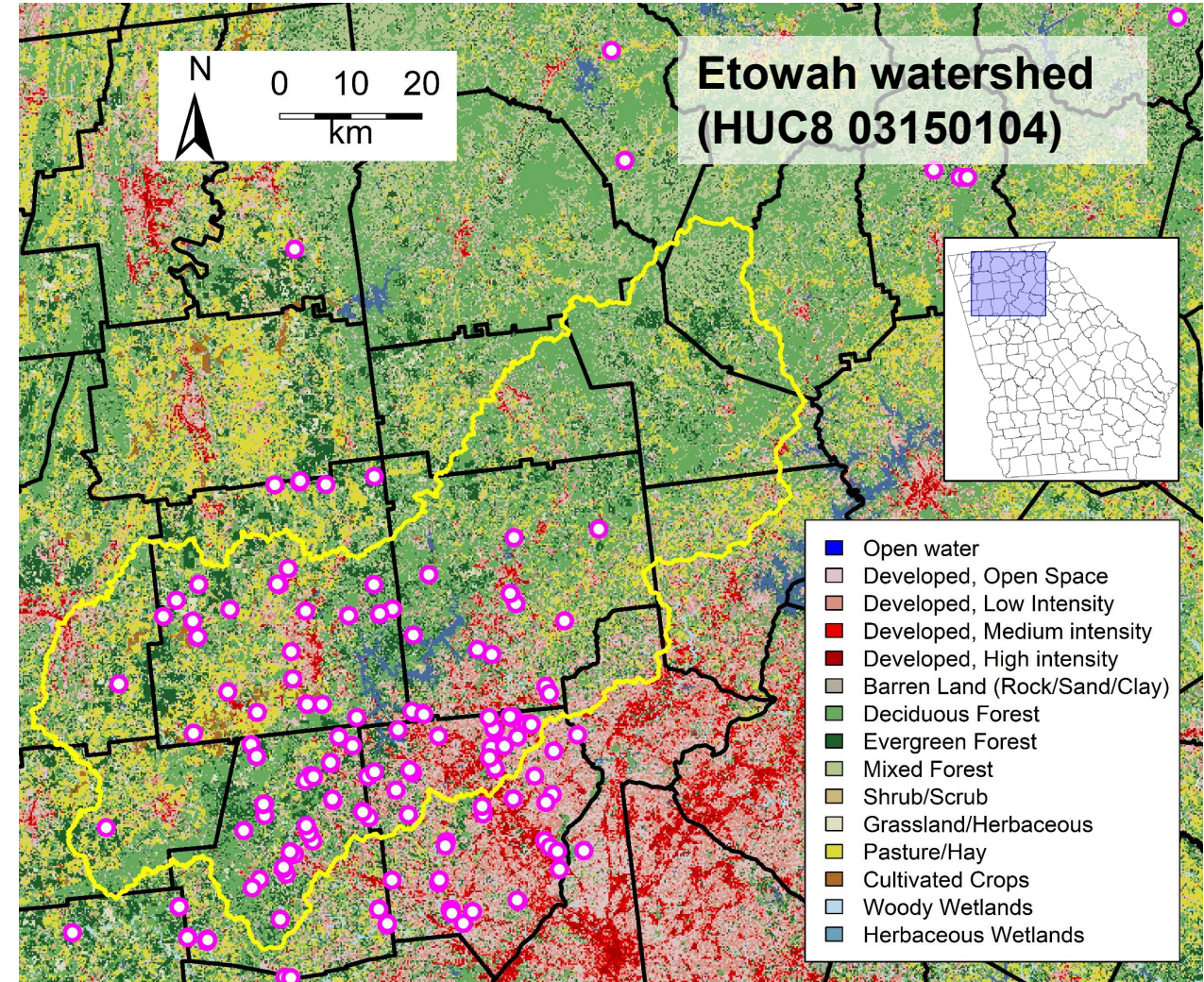
demersal
rocky substrates
guarder and nester

Questions and Hypotheses

- **How do urbanization, land use, and environmental conditions within the Etowah watershed affect its fish communities?**
- **Hypothesis 1:** Anthropogenic land use and human activity within a catchment influence the **biodiversity** of stream fish communities
- **Hypothesis 2:** Anthropogenic land use and human activity within a catchment influence the **life history strategies** of stream fish communities

Methods

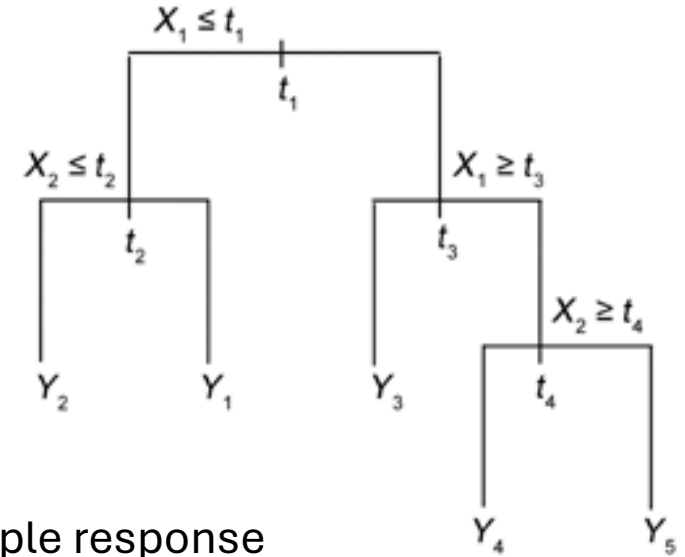
- Stream fish community samples collected from 2000-2018 by Dr. Bill Ensign of Kennesaw State University
 - Standardized methods (IBI protocol)
 - Dataset digitized, cleaned, and taxonomy updated
 - In-stream habitat data not routinely recorded
 - 221 samples
- Environmental variables derived from public data sources
 - Catchments delineated by USGS STREAMSTATS for each location
 - Catchment characteristics derived from NLCD
 - Streamflow estimated using HEC-HMS based on NED, SSURGO, Daymet



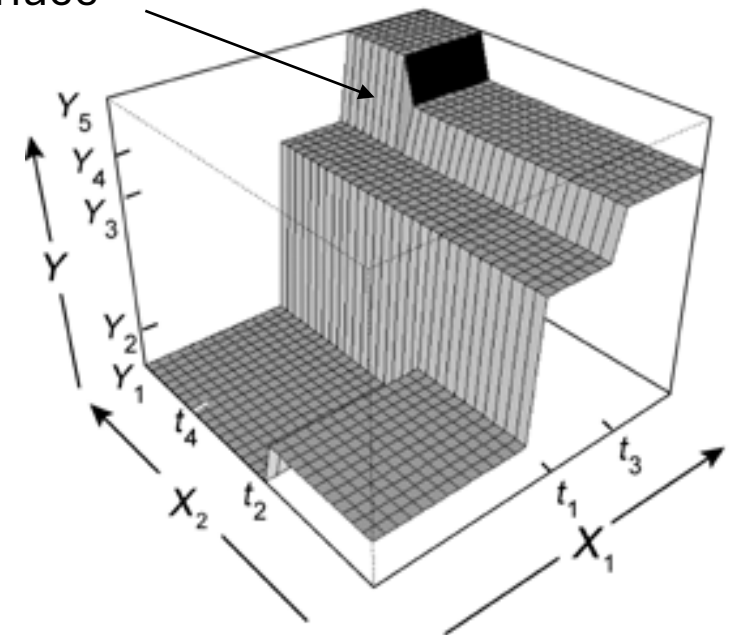
Data analysis

- Hierarchical cluster analysis with Gower distance metric to identify fish ecological strategy groups (ESG)
- Distance-based functional diversity indices to quantify functional diversity in stream samples
- Boosted regression tree analysis to model biodiversity and functional diversity using watershed-scale variables
- All analyses conducted in R version 4.4.1, packages **FD**, **gbm**, **dismo**

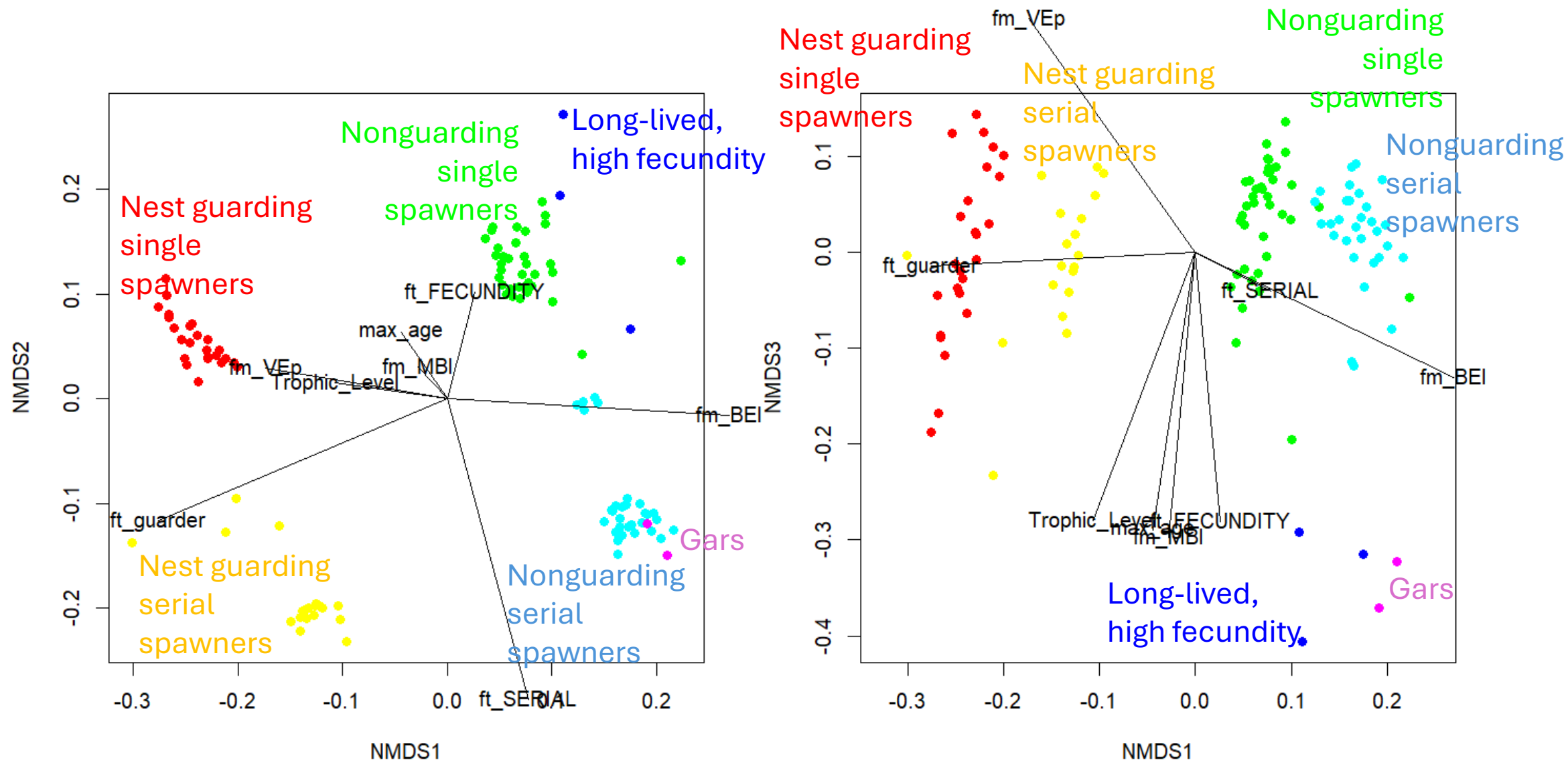
Example tree submodel



Example response surface

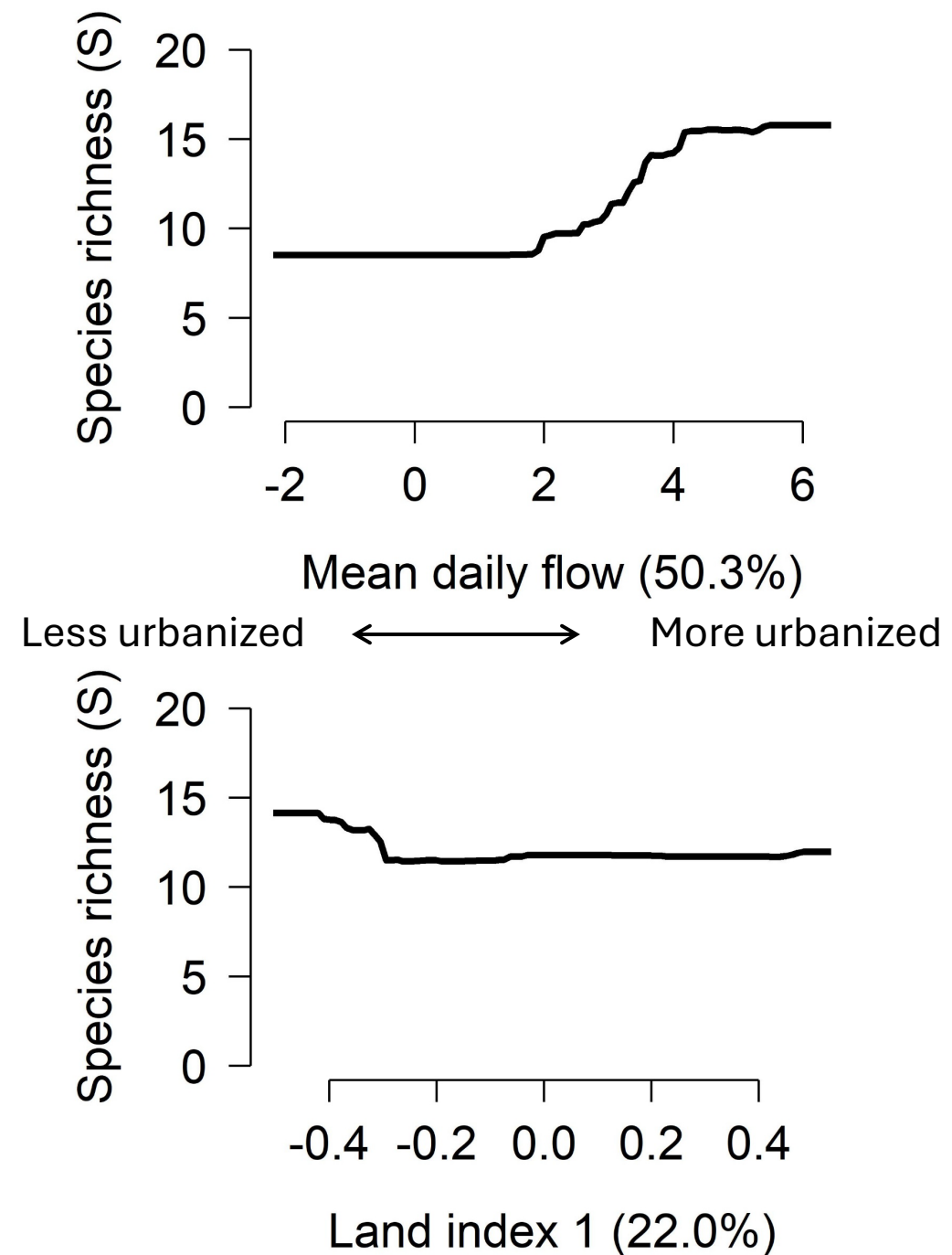


Ecological strategy groups



Results: Fish species richness

- Average flow was the most influential predictor
 - (50.3% of model's explanatory power)
 - Higher flow → higher richness
- Urbanization decreased species richness
- Model explained 39.6% of deviance



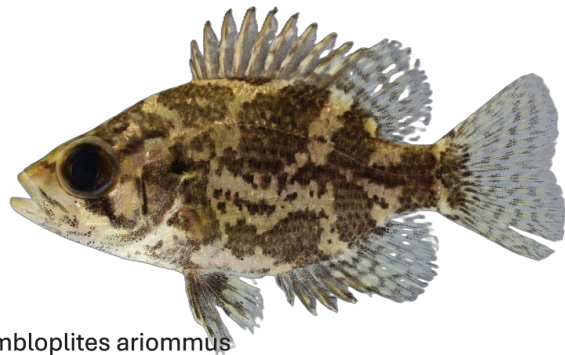
Nest guarding single spawner richness



Cottus bairdii
(mottled sculpin)

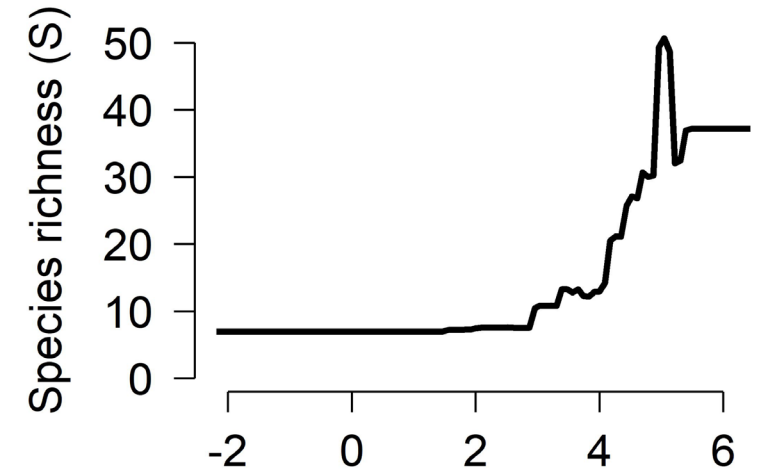


Micropterus coosae
(red eye bass)

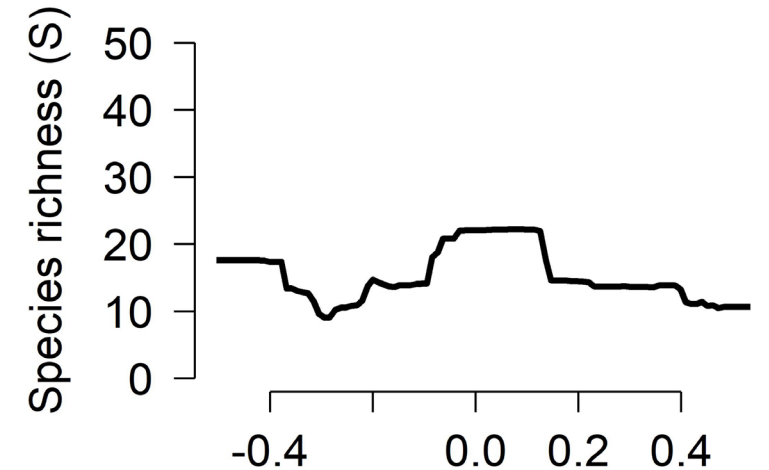


Ambloplites ariommus
(shadow bass)

$Q^2 = 0.177$



Less urbanized ← Mean flow (36.2%) → More urbanized



Land index 1(22.0%)

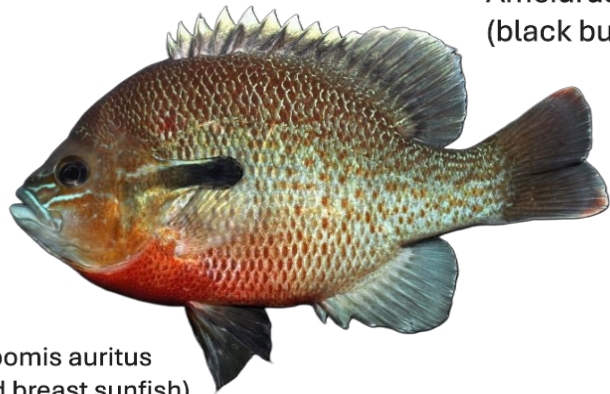
Nest guarding serial spawner richness



Gambusia holbrooki
(eastern mosquitofish)

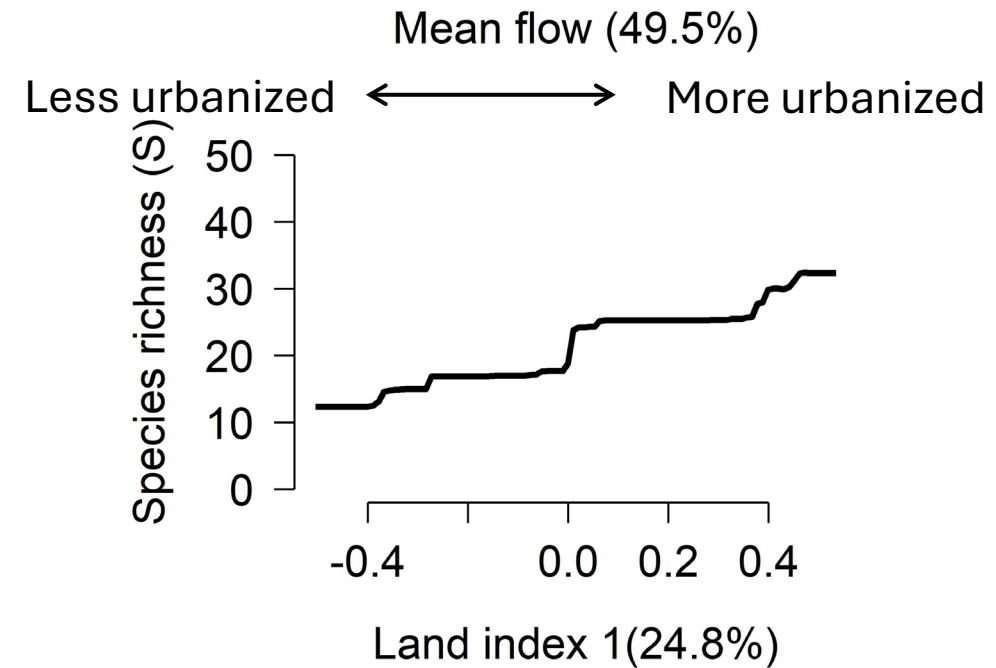
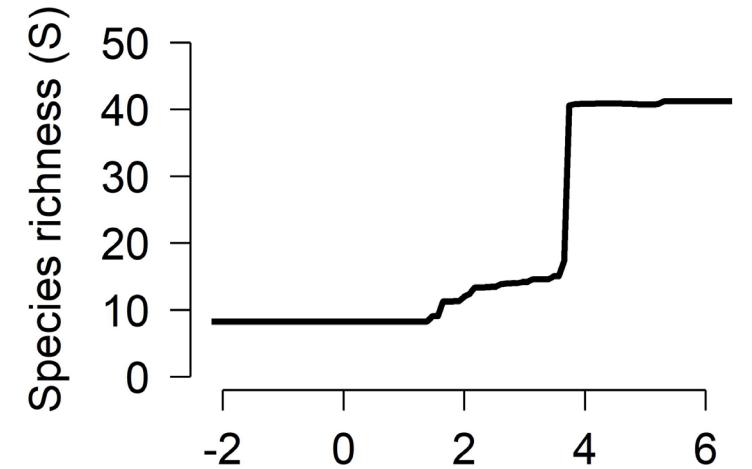


Ameiurus melas
(black bullhead)



Lepomis auritus
(red breast sunfish)

$Q^2 = 0.243$



Nonguarding single spawner richness



Notropis leuciodus
(Tennessee shiner)



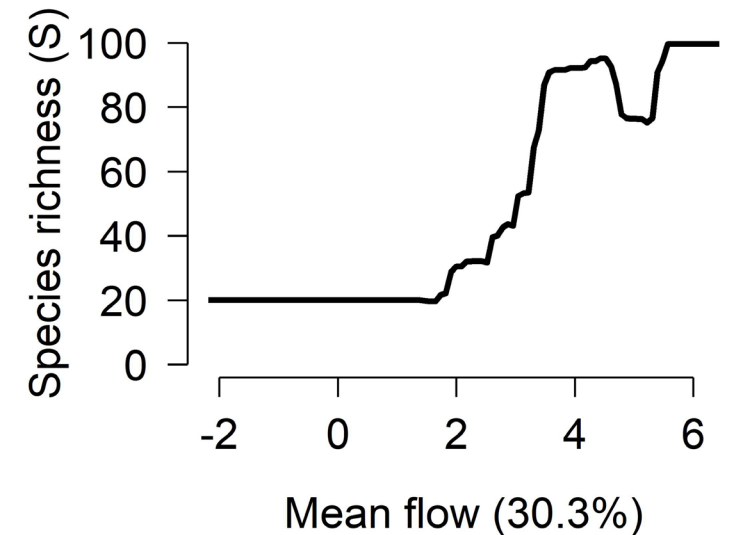
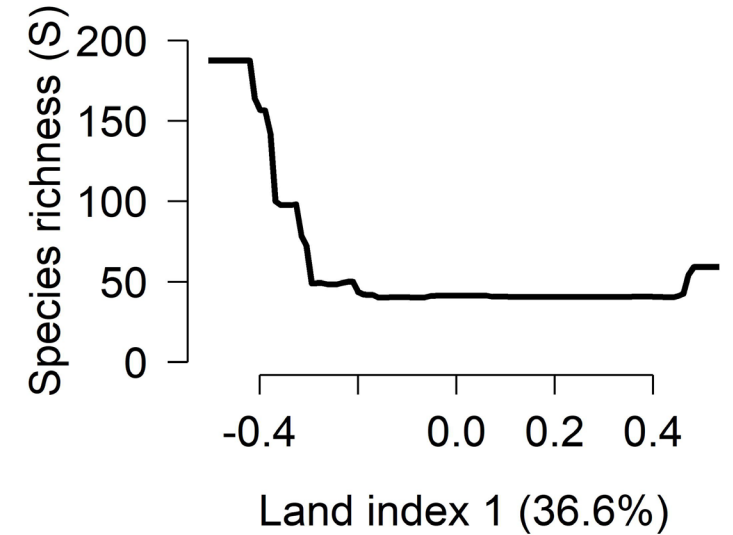
Moxostoma duquesnei
(Black redhorse)

Hypentelium etowanum
(Alabama hogsucker)



$$Q^2 = 0.271$$

Less urbanized ← → More urbanized



Nonguarding serial spawner richness



Cyprinella gibbsi
(Tallapoosa shiner)



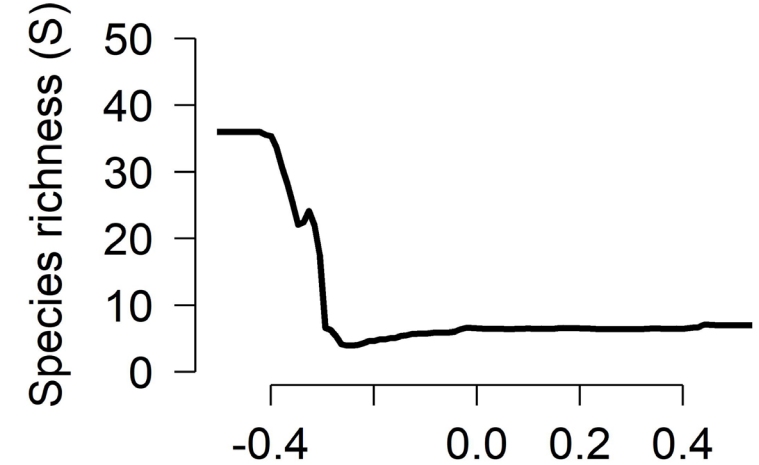
Etheostoma scotti
(Cherokee darter)



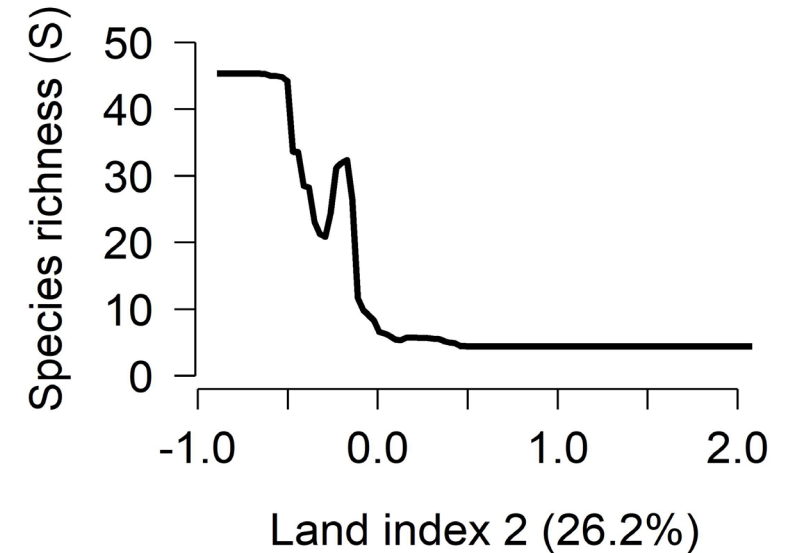
Etheostoma etowahae
(Etowah darter)

$Q^2 = 0.455$

Less urbanized ← → More urbanized



Land index 1 (36.6%)
More forested ← → More open

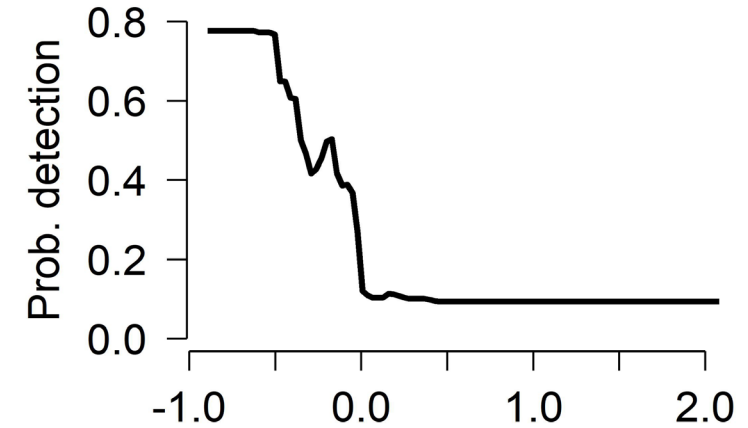


Cherokee darter (*Etheostoma scotti*) presence

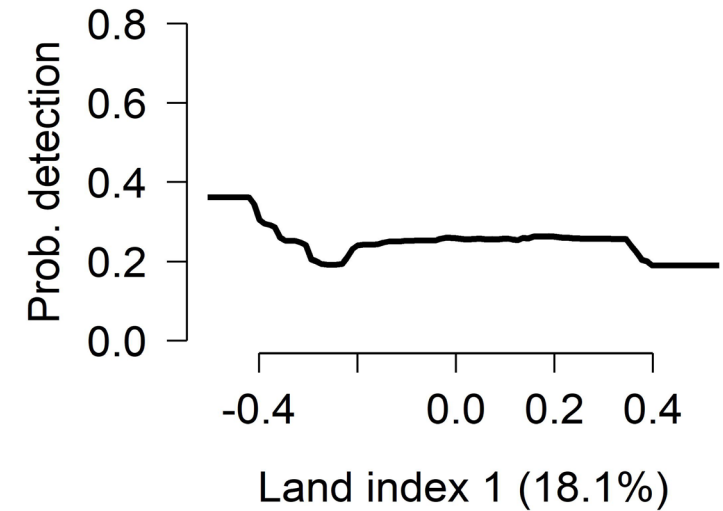


$Q^2 = 0.293$; AUC = 0.945

More forested ← → More open



Less urbanized ← → More urbanized

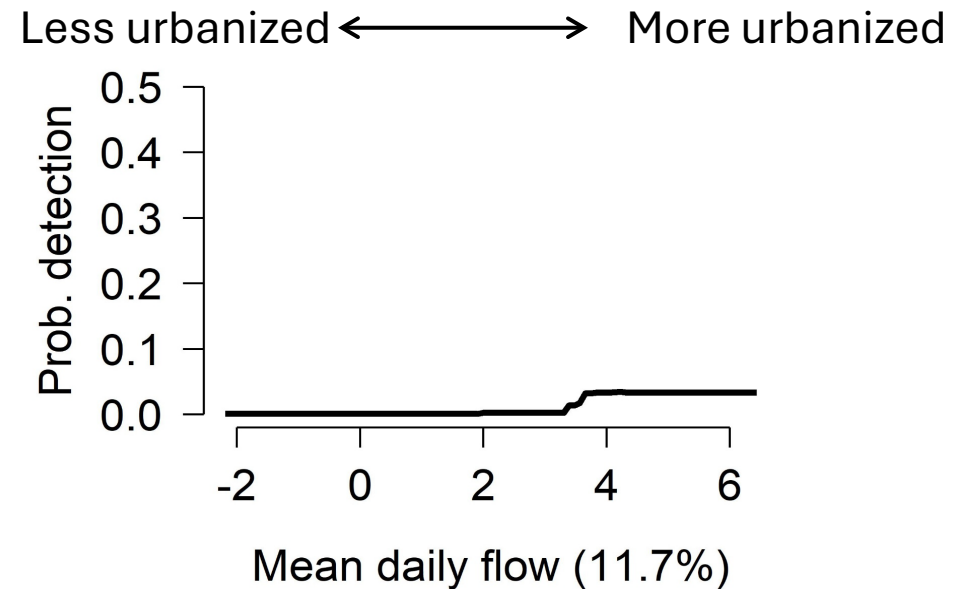
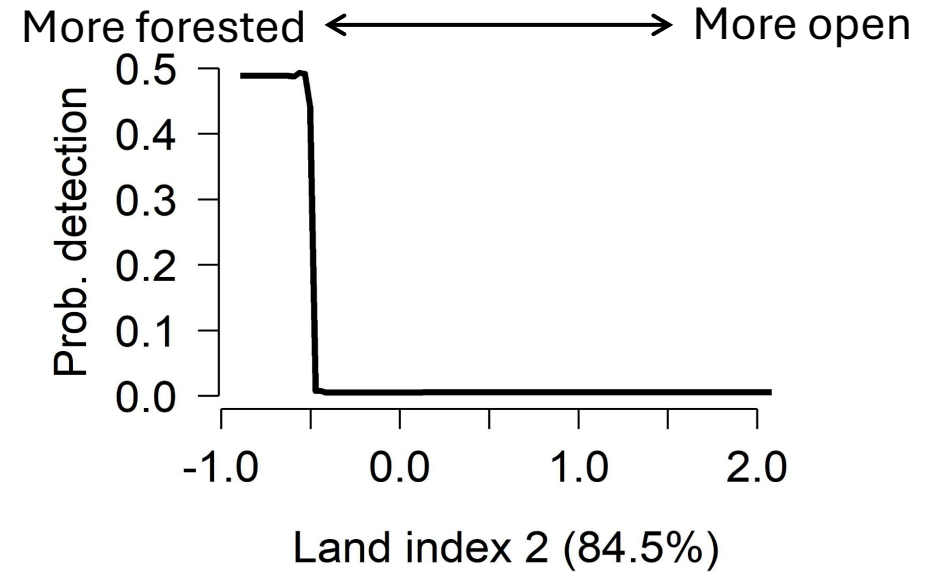


Etowah darter (*Etheostoma etowahae*) presence



Etheostoma etowahae
(Etowah darter)

$Q^2 = 0.633$; AUC = 0.989

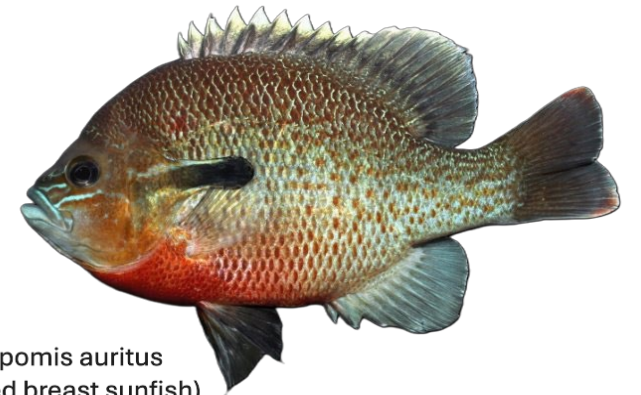


Discussion

- **Hypothesis 1: supported** because of influence of urbanization
- **Hypothesis 2: not supported** because mix of life history strategies not well-explained by predictors



Notropis leuciodus
(Tennessee shiner)



Lepomis auritus
(red breast sunfish)

Conclusions

- Streamflow, forest cover, and urbanization consistently among most influential predictors of fish biodiversity
- Streamflow variability not influential
- Relative influence of different predictors varied by species and functional group
- T&E species (*E. etowahae* and *E. scotti*) most sensitive to forest loss



Thank you!



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